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Grade	**85.50** out of 100.00

Question 1

Correct

Mark 2.00 out of 2.00

The following loop displays _____.

```
for (int i = 1; i <= 10; i++) {
  System.out.print(i + " ");
  i++;
}
```

Select one:

- A. 1 2 3 4 5 6 7 8 9
- B. 1 2 3 4 5 6 7 8 9 10
- C. 1 2 3 4 5
- D. 1 3 5 7 9 ✔ In this for loop, i is initialized to 1. i++ adds 1 to i at the end of the iteration. In the action-after-each-iteration, i is again incremented by 1. So, i is in effect incremented twice. So, i is 1, then 3, 5, 7, and 9. The correct answer is D.
- E. 2 4 6 8 10

In this for loop, i is initialized to 1. i++ adds 1 to i at the end of the iteration. In the action-after-each-iteration, i is again incremented by 1. So, i is in effect incremented twice. So, i is 1, then 3, 5, 7, and 9. The correct answer is D.

In this for loop, i is initialized to 1. i++ adds 1 to i at the end of the iteration. In the action-after-each-iteration, i is again incremented by 1. So, i is in effect incremented twice. So, i is 1, then 3, 5, 7, and 9. The correct answer is D.

The correct answer is: 1 3 5 7 9

Question 2

Correct

Mark 1.00 out of 1.00

Suppose i is an int type variable. Which of the following statements display the character whose Unicode is stored in variable i?

Select one:

- A. System.out.println((char)i); ✔ (char)i casts a number into a character.
- B. System.out.println(i + " ");
- C. System.out.println((int)i);
- D. System.out.println(i);

(char)i casts a number into a character.

(char)i casts a number into a character.

The correct answer is: System.out.println((char)i);

Question 3

Correct

Mark 1.00 out of 1.00

Suppose x is a char variable with a value 'b'. What is the output of the statement System.out.println(++x)?

Select one:

- A. a
- B. b
- C. c ✔ The ++ and -- operators can be applied to a char variable. ++x is preincrement. x is 'b' before ++x. After ++x, x becomes c.
- D. d

The ++ and -- operators can be applied to a char variable. ++x is preincrement. x is 'b' before ++x. After ++x, x becomes c.

The ++ and -- operators can be applied to a char variable. ++x is preincrement. x is 'b' before ++x. After ++x, x becomes c.

The correct answer is: c

Question 4

Correct

Mark 2.00 out of 2.00

A Java character is stored in _____.

Select one:

- A. one byte
- B. two bytes ✔ Java characters use Unicode encoding.
- C. three bytes
- D. four bytes

Java characters use Unicode encoding.

Java characters use Unicode encoding.

The correct answer is: two bytes

Question 5

Partially correct

Mark 0.50 out of 1.00

Analyze the following code. Please select all that apply.

```
public class Test {
public static void main (String[] args) {
   int i = 0;
   for (i = 0; i < 10; i++);
    System.out.println(i + 4);
  }
}
```

Select one or more:

- A. The program compiles despite the semicolon (;) on the for loop line, and displays 14.
- B. The program has a compile error because of the semicolon (;) on the for loop line.
- C. The for loop in this program is same as for (i = 0; i < 10; i++) { }; System.out.println(i + 4); ✔ This is a logic error. System.out.println(i + 4) is not a part the for loop because the for loop ends with the last semicolon at for (i=0; i < 10; i++);
- D. The program compiles despite the semicolon (;) on the for loop line, and displays 4. ✖

This is a logic error. System.out.println(i + 4) is not a part the for loop because the for loop ends with the last semicolon at for (i=0; i < 10; i++);

The correct answers are: The program compiles despite the semicolon (;) on the for loop line, and displays 14., The for loop in this program is same as for (i = 0; i < 10; i++) { }; System.out.println(i + 4);

Question 6

Correct

Mark 2.00 out of 2.00

Which of the following loops correctly computes 1/2 + 2/3 + 3/4 + ... + 99/100?

A:
```
double sum = 0;
for (int i = 1; i <= 99; i++) {
  sum = i / (i + 1);
}
System.out.println("Sum is " + sum);
```

B:
```
double sum = 0;
for (int i = 1; i < 99; i++) {
  sum += i / (i + 1);
}
System.out.println("Sum is " + sum);
```

C:
```
double sum = 0;
for (int i = 1; i <= 99; i++) {
  sum += 1.0 * i / (i + 1);
}
System.out.println("Sum is " + sum);
```

D:
```
double sum = 0;
for (int i = 1; i <= 99; i++) {
  sum += i / (i + 1.0);
}
System.out.println("Sum is " + sum);
```

E:
```
double sum = 0;
for (int i = 1; i < 99; i++) {
  sum += i / (i + 1.0);
}
System.out.println("Sum is " + sum);
```

Select one:

- A. BCD
- B. ABCD
- C. B
- D. CDE
- E. CD ✔

Note that 1 / 2 is 0.5 in a math expression. In Java, however, integer division yields an integer. The fraction part is truncated. Therefore, i / (i + 1) is 0. A and B are incorrect. E is incorrect because the last i in the loop is 98. So, the last item 99 / 100.0 is not added to sum. So, the correct answer is CD.

Note that 1 / 2 is 0.5 in a math expression. In Java, however, integer division yields an integer. The fraction part is truncated. Therefore, i / (i + 1) is 0. A and B are incorrect. E is incorrect because the last i in the loop is 98. So, the last item 99 / 100.0 is not added to sum. So, the correct answer is CD.

The correct answer is: CD

Question 7

Correct

Mark 2.00 out of 2.00

To check if a string s contains the suffix "Java", you may write _____. Please select all that apply.

Select one or more:

- A. if (s.substring(s.length() - 4).equals("Java")) ... ✔ s.lastIndexOf("Java") >= 0 does not indicate that Java is the suffix of the string.
- B. if (s.endsWith("Java")) ... ✔ s.lastIndexOf("Java") >= 0 does not indicate that Java is the suffix of the string.
- C. if (s.charAt(s.length() - 4) == 'J' && s.charAt(s.length() - 3) == 'a' && s.charAt(s.length() - 2) == 'v' && s.charAt(s.length() - 1) == 'a') ... ✔ s.lastIndexOf("Java") >= 0 does not indicate that Java is the suffix of the string.
- D. if (s.lastIndexOf("Java") >= 0) ... ✖
- E. if (s.substring(s.length() - 5).equals("Java")) ...

s.lastIndexOf("Java") >= 0 does not indicate that Java is the suffix of the string.

s.lastIndexOf("Java") >= 0 does not indicate that Java is the suffix of the string.

The correct answers are: if (s.endsWith("Java")) ..., if (s.substring(s.length() - 4).equals("Java")) ..., if (s.charAt(s.length() - 4) == 'J' && s.charAt(s.length() - 3) == 'a' && s.charAt(s.length() - 2) == 'v' && s.charAt(s.length() - 1) == 'a') ...

Question 8

Correct

Mark 2.00 out of 2.00

Will System.out.println((char)4) display 4?

Select one:

- A. Yes
- B. No ✔ The character whose Unicode is \u0004 is to be displayed, not number 4.

The character whose Unicode is \u0004 is to be displayed, not number 4.

The character whose Unicode is \u0004 is to be displayed, not number 4.

The correct answer is: No

Question 9

Correct

Mark 1.00 out of 1.00

Note that the Unicode for character A is 65. The expression 'A' + 1 evaluates to _____.

Select one:

- A. 66 ✔ When a character adds with a number, the character is converted into a int. The correct answer is A.
- B. B
- C. A1
- D. Illegal expression

When a character adds with a number, the character is converted into a int. The correct answer is A.

When a character adds with a number, the character is converted into a int. The correct answer is A.

The correct answer is: 66

Question 10

Correct

Mark 2.00 out of 2.00

The statement System.out.printf("%10s", 123456) outputs _____. (Note: * represents a space)

Select one:

- A. 123456****
- B. 23456*****
- C. 12345*****
- D. ****123456 ✔ %10s specifies to display a string with width 10. By default, it is right justified. So, the correct answer is D.

%10s specifies to display a string with width 10. By default, it is right justified. So, the correct answer is D.

%10s specifies to display a string with width 10. By default, it is right justified. So, the correct answer is D.

The correct answer is: ****123456

Question 11

Correct

Mark 2.00 out of 2.00

What balance after the following code is executed?

```
int balance = 10;

while (balance >= 1) {
  if (balance < 9)
    continue;
  balance = balance - 9;
}
```

Select one:

- A. -1
- B. 0
- C. 1
- D. 2
- E. The loop does not end ✔

Before the loop, balance is 10. The loop-continuation-condition is true (10 >= 1). In the first iteration, balance is reduced to 1. Since 1 >= 1 is true, the loop body is executed. Since balance < 9 is true, the continue statement is executed to skip the rest statement in the iteration. balance continues to be 1 and the continue statement is executed to skip the rest statement in the iteration. So, the loop runs infinitely. So, the correct answer to this question is E.

Before the loop, balance is 10. The loop-continuation-condition is true (10 >= 1). In the first iteration, balance is reduced to 1. Since 1 >= 1 is true, the loop body is executed. Since balance < 9 is true, the continue statement is executed to skip the rest statement in the iteration. balance continues to be 1 and the continue statement is executed to skip the rest statement in the iteration. So, the loop runs infinitely. So, the correct answer to this question is E.

Before the loop, balance is 10. The loop-continuation-condition is true (10 >= 1). In the first iteration, balance is reduced to 1. Since 1 >= 1 is true, the loop body is executed. Since balance < 9 is true, the continue statement is executed to skip the rest statement in the iteration. balance continues to be 1 and the continue statement is executed to skip the rest statement in the iteration. So, the loop runs infinitely. So, the correct answer to this question is E.

The correct answer is: The loop does not end

Question 12

Correct

Mark 2.00 out of 2.00

Will the following program terminate?

```
int balance = 10;

while (true) {
  if (balance &amp;lt; 9)
    break;
  balance = balance - 9;
}
```

Select one:

- A. Yes ✔ Yes. Before the loop, balance is 10. The loop-continuation-condition is always true. In the first iteration, balance is reduced to 1. In the second iteration, the break statement is executed since (balance < 9) is now true. The correct answer for this question is A.
- B. No

Yes. Before the loop, balance is 10. The loop-continuation-condition is always true. In the first iteration, balance is reduced to 1. In the second iteration, the break statement is executed since (balance < 9) is now true. The correct answer for this question is A.

Yes. Before the loop, balance is 10. The loop-continuation-condition is always true. In the first iteration, balance is reduced to 1. In the second iteration, the break statement is executed since (balance < 9) is now true. The correct answer for this question is A.

The correct answer is: Yes

Question 13

Correct

Mark 2.00 out of 2.00

Math.sin(Math.PI) returns _____.

Select one:

- A. 0.0 ✔ Note that Math.PI is 180 degrees.
- B. 0.4
- C. 0.5
- D. 1.0

Note that Math.PI is 180 degrees.

Note that Math.PI is 180 degrees.

The correct answer is: 0.0

Question 14

Partially correct

Mark 1.00 out of 2.00

Which of the following assignment statements is correct? Please select all that apply.

Select one or more:

- A. char c = 'd'; ✔ Choice B is also correct, because an int value can be implicitly cast into a char variable. The Unicode of the character is the int value. In this case, the character is d (see Appendix B).
- B. char c = 100;
- C. char c = "d";
- D. char c = "100"; ✖

Choice B is also correct, because an int value can be implicitly cast into a char variable. The Unicode of the character is the int value. In this case, the character is d (see Appendix B).

The correct answers are: char c = 'd';, char c = 100;

Question 15

Correct

Mark 2.00 out of 2.00

'3' - '2' + 'm' / 'n' is _____.

Select one:

- A. 0
- B. 1 ✔ When an operand is a character in an arithmetic expression, the character is casted to an int value.
- C. 2
- D. 3

When an operand is a character in an arithmetic expression, the character is casted to an int value.

When an operand is a character in an arithmetic expression, the character is casted to an int value.

The correct answer is: 1

Question 16

Correct

Mark 2.00 out of 2.00

Which of the following is correct to obtain a random integer between 5 and 10?

Select one:

- A. 5 + Math.random() * 6
- B. 5 + (int) (Math.random() * 6) ✔ In A, 5 + Math.random() * 6 returns a double value from 5.0 to 11.0 (excluding 11.0). In C, 5 + Math.random() * 5 returns a double value from 5.0 to 10.0 (excluding 10.0). In D, 5 + (int)(Math.random() * 5) returns an integer from 5 to 10. B is correct.
- C. 5 + Math.random() * 5
- D. 5 + (int)(Math.random() * 5)

In A, 5 + Math.random() * 6 returns a double value from 5.0 to 11.0 (excluding 11.0). In C, 5 + Math.random() * 5 returns a double value from 5.0 to 10.0 (excluding 10.0). In D, 5 + (int)(Math.random() * 5) returns an integer from 5 to 10. B is correct.

In A, 5 + Math.random() * 6 returns a double value from 5.0 to 11.0 (excluding 11.0). In C, 5 + Math.random() * 5 returns a double value from 5.0 to 10.0 (excluding 10.0). In D, 5 + (int)(Math.random() * 5) returns an integer from 5 to 10. B is correct.

The correct answer is: 5 + (int)(Math.random() * 6)

Question 17

Correct

Mark 2.00 out of 2.00

To check whether a char variable ch is an uppercase letter, you write _____.

Select one:

- A. (ch >= 'A' && ch >= 'Z')
- B. (ch >= 'A' && ch <= 'Z') ✔ A is wrong because ch >= 'Z'. C is wrong because of using ||. D is wrong because of incorrect syntax. The correct answer is B.
- C. (ch >= 'A' || ch <= 'Z')
- D. ('A' <= ch <= 'Z')

A is wrong because ch >= 'Z'. C is wrong because of using ||. D is wrong because of incorrect syntax. The correct answer is B.

A is wrong because ch >= 'Z'. C is wrong because of using ||. D is wrong because of incorrect syntax. The correct answer is B.

The correct answer is: (ch >= 'A' && ch <= 'Z')

Question 18

Correct

Mark 1.00 out of 1.00

Suppose s is a string with the value "java". What will be assigned to x if you execute the following code?

```
char x = s.charAt(4);
```

Select one:

- A. 'a'
- B. 'v'
- C. Nothing will be assigned to x, because the execution causes the runtime error StringIndexOutofBoundsException. ✔ The string index starts from 0 and the last index is s.length() - 1. s.charAt(4) is out of bounds.

The string index starts from 0 and the last index is s.length() - 1. s.charAt(4) is out of bounds.

The string index starts from 0 and the last index is s.length() - 1. s.charAt(4) is out of bounds.

The correct answer is: Nothing will be assigned to x, because the execution causes the runtime error StringIndexOutofBoundsException.

Question 19

Correct

Mark 1.00 out of 1.00

How many times will the following code print "Welcome to Java"?

```
int count = 0;
do {
  System.out.println("Welcome to Java");
} while (count++ < 10);
```

Select one:

- A. 8
- B. 9
- C. 10
- D. 11 ✔ The count is initialized to 0 before the loop. The loop is executed for the first time when count is 0. (count++ < 10) increments count by 1 and uses the old count value to check if count < 10. So, the loop is executed 11 times for count from 0 to 10. Note that when count is 10, the loop body is executed before the loop is exited. The correct answer is D.
- E. 0

The count is initialized to 0 before the loop. The loop is executed for the first time when count is 0. (count++ < 10) increments count by 1 and uses the old count value to check if count < 10. So, the loop is executed 11 times for count from 0 to 10. Note that when count is 10, the loop body is executed before the loop is exited. The correct answer is D.

The count is initialized to 0 before the loop. The loop is executed for the first time when count is 0. (count++ < 10) increments count by 1 and uses the old count value to check if count < 10. So, the loop is executed 11 times for count from 0 to 10. Note that when count is 10, the loop body is executed before the loop is exited. The correct answer is D.

The correct answer is: 11

Question 20

Correct

Mark 1.00 out of 1.00

The Unicode of 'a' is 97. What is the Unicode for 'c'?

Select one:

- A. 96
- B. 97
- C. 98
- D. 99 ✔ The Unicode for letters and numbers are allocated in a natural order. So, b is after a and c is after b, and so on.

The Unicode for letters and numbers are allocated in a natural order. So, b is after a and c is after b, and so on.

The Unicode for letters and numbers are allocated in a natural order. So, b is after a and c is after b, and so on.

The correct answer is: 99

Question 21

Correct

Mark 2.00 out of 2.00

Which pattern is produced by the following code?

```
for (int i = 1; i <= 6; i++) {
  for (int j = 6; j >= 1; j--)
   System.out.print(j <= i ? j + " " : " " + " ");
  System.out.println();
}
```

Pattern A	Pattern B	Pattern C	Pattern D
1	123456	1	123456
12	12345	21	12345
123	1234	321	1234
1234	123	4321	123
12345	12	54321	12
123456	1	654321	1

Select one:

- A. Pattern A
- B. Pattern B
- C. Pattern C ✔ The outer loop displays a line. The inner loop displays all the numbers in the line. The outer loop is repeated 6 times for i from 1 to 6. For each i, the inner loop displays i as the first number in the line since j is from 6 to 1. j is printed for j <= i. Otherwise, a space is printed. Clearly, this code will print Pattern C.
- D. Pattern D

The outer loop displays a line. The inner loop displays all the numbers in the line. The outer loop is repeated 6 times for i from 1 to 6. For each i, the inner loop displays i as the first number in the line since j is from 6 to 1. j is printed for j <= i. Otherwise, a space is printed. Clearly, this code will print Pattern C.

The outer loop displays a line. The inner loop displays all the numbers in the line. The outer loop is repeated 6 times for i from 1 to 6. For each i, the inner loop displays i as the first number in the line since j is from 6 to 1. j is printed for j <= i. Otherwise, a space is printed. Clearly, this code will print Pattern C.

The correct answer is: Pattern C

Question 22

Correct

Mark 2.00 out of 2.00

What is Math.ceil(3.6)?

Select one:

- A. 3.0
- B. 3
- C. 4.0 ✔ Note that ceil returns a double value.
- D. 5.0

Note that ceil returns a double value.

Note that ceil returns a double value.

The correct answer is: 4.0

Question 23

Correct

Mark 2.00 out of 2.00

Which of the following statements prints smith\exam1\test.txt?

Select one:

- A. System.out.println("smith\exam1\test.txt");
- B. System.out.println("smith"\exam1"\test.txt");
- C. System.out.println("smith\"exam1\"test.txt");
- D. System.out.println("smith\\exam1\\test.txt"); ✔ To represent the \ character, use \\, because it is an escape character.

To represent the \ character, use \\, because it is an escape character.

To represent the \ character, use \\, because it is an escape character.

The correct answer is: System.out.println("smith\\exam1\\test.txt");

Question 24

Incorrect

Mark 0.00 out of 2.00

Analyze the following code.

```
double sum = 0;
for (double d = 0; d < 10; sum += sum + d) {
  d += 0.1;
}
```

Select one:

- A. The program has a syntax error because the adjustment statement is incorrect in the for loop. ✖
- B. The program has a syntax error because the control variable in the for loop cannot be of the double type.
- C. The program compiles but does not stop because d would always be less than 10.
- D. The program compiles and runs fine.

In this loop, the loop initial action is d = 0, the continuation condition is d < 10, and the action-after-each-iteration is sum += sum + d. The loop body is d += 1. This code is correct in syntax. d is initially 0 and d += 0.1 adds 0.1 to d in each iteration. Eventually d will be greater than or equal to 10. So, the loop will terminate. The correct answer for this question is D) The program compiles and runs fine.

The correct answer is: The program compiles and runs fine.

Question 25

Correct

Mark 2.00 out of 2.00

Which of the following is the correct statement to return JAVA?

Select one:

- A. toUpperCase("Java")
- B. "Java".toUpperCase("Java")
- C. "Java".toUpperCase() ✔ The correct method is toUpperCase(). So, C is correct.
- D. String.toUpperCase("Java")

The correct method is toUpperCase(). So, C is correct.

The correct method is toUpperCase(). So, C is correct.

The correct answer is: "Java".toUpperCase()

Question 26

Correct

Mark 2.00 out of 2.00

What is Math.floor(3.6)?

Select one:

- A. 3.0 ✔ Note that floor returns a double value.
- B. 3
- C. 4
- D. 5.0

Note that floor returns a double value.

Note that floor returns a double value.

The correct answer is: 3.0

Question 27

Partially correct

Mark 0.50 out of 1.00

An int variable can hold _____. Please select all that apply.

Select one or more:

- A. 'x'
- B. 120 ✔ Choice A is also correct, because a character can be implicitly cast into an int variable. The Unicode value of character is assignment to the int variable. In this case, the code is 120 (see Appendix B).
- C. 120.0
- D. "x"
- E. "120" ✖

Choice A is also correct, because a character can be implicitly cast into an int variable. The Unicode value of character is assignment to the int variable. In this case, the code is 120 (see Appendix B).

The correct answers are: 'x', 120

Question 28

Correct

Mark 2.00 out of 2.00

How many times will the following code print "Welcome to Java"?

```
int count = 0;
do {
  System.out.println("Welcome to Java");
} while (++count < 10);
```

Select one:

- A. 8
- B. 9
- C. 10 ✔ The count is initialized to 0 before the loop. The loop is executed for the first time when count is 0. (++count < 10) increments count by 1 and uses the new count value to check if count < 10. So, the loop is executed 10 times for count from 0 to 9. Note that when count is 9, ++count becomes 10, the loop exits. The correct answer is C.
- D. 11
- E. 0

The count is initialized to 0 before the loop. The loop is executed for the first time when count is 0. (++count < 10) increments count by 1 and uses the new count value to check if count < 10. So, the loop is executed 10 times for count from 0 to 9. Note that when count is 9, ++count becomes 10, the loop exits. The correct answer is C.

The count is initialized to 0 before the loop. The loop is executed for the first time when count is 0. (++count < 10) increments count by 1 and uses the new count value to check if count < 10. So, the loop is executed 10 times for count from 0 to 9. Note that when count is 9, ++count becomes 10, the loop exits. The correct answer is C.

The correct answer is: 10

Question 29

Correct

Mark 2.00 out of 2.00

Suppose the input for number is 9. What is the output from running the following program?

```
import java.util.Scanner;

public class Test {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an integer: ");
    int number = input.nextInt();

    int i;

    boolean isPrime = true;
    for (i = 2; i < number && isPrime; i++) {
      if (number % i == 0) {
        isPrime = false;
      }
    }

    System.out.println("i is " + i);

    if (isPrime)
      System.out.println(number + " is prime");
    else
      System.out.println(number + " is not prime");
  }
}
```

Select one:

- A. i is 3 followed by 9 is prime
- B. i is 3 followed by 9 is not prime
- C. i is 4 followed by 9 is prime
- D. i is 4 followed by 9 is not prime ✔

The input number is 9. isPrime is initialized to true. The loop tests if number is divisible by i for i from 2, 3, and so on. When i is 3, isPrime is false. The loop continuation condition becomes false. The loop is not finished. Before the loop continuation condition is checked, i++ increments i by 1. So, output is that i is 4 and following by 9 is not prime. So, the correct answer is executed n times for i from 1 to n. So, the correct answer is D.

The input number is 9. isPrime is initialized to true. The loop tests if number is divisible by i for i from 2, 3, and so on. When i is 3, isPrime is false. The loop continuation condition becomes false. The loop is not finished. Before the loop continuation condition is checked, i++ increments i by 1. So, output is that i is 4 and following by 9 is not prime. So, the correct answer is executed n times for i from 1 to n. So, the correct answer is D.

The correct answer is: i is 4 followed by 9 is not prime

Question 30

Correct

Mark 2.00 out of 2.00

Note that the Unicode for character A is 65. The expression "A" + 1 evaluates to _____.

Select one:

- A. 66
- B. B
- C. A1 ✔ When a string adds with a number, the number is converted into a string. The correct answer is C.
- D. Illegal expression

When a string adds with a number, the number is converted into a string. The correct answer is C.

When a string adds with a number, the number is converted into a string. The correct answer is C.

The correct answer is: A1

Question 31

Correct

Mark 2.00 out of 2.00

What is the output of System.out.println('z' - 'a')?

Select one:

- A. 25 ✔ The Unicode offset between z and a is 25.
- B. a
- C. 26
- D. z

The Unicode offset between z and a is 25.

The Unicode offset between z and a is 25.

The correct answer is: 25

Question 32

Incorrect

Mark 0.00 out of 2.00

What is y after the following for loop statement is executed?

```
int y = 0;
for (int i = 0; i < 10; ++i) {
  y += 1;
}
```

Select one:

- A. 9
- B. 10
- C. 11 ✖
- D. 12

Before the loop, y is 0. The loop is executed 10 times. Each time, 1 is added to y. So, after the loop is finished, y is 10. The correct answer is B.

The correct answer is: 10

Question 33

Incorrect

Mark 0.00 out of 1.00

The _____ method parses a string s to a double value.

Select one:

- A. double.parseDouble(s); ✖
- B. Double.parsedouble(s);
- C. double.parse(s);
- D. Double.parseDouble(s);

The parseDouble method is defined in the Double class. D is correct.

The correct answer is: Double.parseDouble(s);

Question 34

Correct

Mark 2.00 out of 2.00

The statement System.out.printf("%3.1f", 1234.56) outputs _____.

Select one:

- A. 123.4
- B. 123.5
- C. 1234.56
- D. 1234.6 ✔ .1 specifies one digit after the decimal point. The rest is rounded up. So, 1234.56 is displayed 1234.6.
- E. 1234.5

.1 specifies one digit after the decimal point. The rest is rounded up. So, 1234.56 is displayed 1234.6.

.1 specifies one digit after the decimal point. The rest is rounded up. So, 1234.56 is displayed 1234.6.

The correct answer is: 1234.6

Question 35

Correct

Mark 2.00 out of 2.00

_____ returns true. Please select all that apply.

Select one or more:

- A. "peter".compareToIgnoreCase("Peter")
- B. "peter".compareToIgnoreCase("peter") ✖
- C. "peter".equalsIgnoreCase("Peter") ✔ The compareToIgnoreCase return an int. So, A and B are wrong. Ignoring case, C, D, and E all return true.
- D. "peter".equalsIgnoreCase("peter") ✔ The compareToIgnoreCase return an int. So, A and B are wrong. Ignoring case, C, D, and E all return true.
- E. "peter".equals("peter") ✔ The compareToIgnoreCase return an int. So, A and B are wrong. Ignoring case, C, D, and E all return true.

The compareToIgnoreCase return an int. So, A and B are wrong. Ignoring case, C, D, and E all return true.

The compareToIgnoreCase return an int. So, A and B are wrong. Ignoring case, C, D, and E all return true.

The correct answers are: "peter".equalsIgnoreCase("Peter"), "peter".equalsIgnoreCase("peter"), "peter".equals("peter")

Question 36

Correct

Mark 2.00 out of 2.00

What is the number of iterations in the following loop?

```
for (int i = 1; i < n; i++) {
  // iteration
}
```

Select one:

- A. 2*n
- B. n
- C. n - 1 ✔ The loop is executed n - 1 times for i from 1 to n - 1. So, the correct answer is C.
- D. n + 1

The loop is executed n - 1 times for i from 1 to n - 1. So, the correct answer is C.

The loop is executed n - 1 times for i from 1 to n - 1. So, the correct answer is C.

The correct answer is: n - 1

Question 37

Correct

Mark 2.00 out of 2.00

Which of the following is correct to obtain the min of x, y, z? Please select all that apply.

Select one or more:

- A. Math.min(x, Math.min(y, z)) ✔ All choices are correct.
- B. Math.min(Math.min(x, y), z) ✔ All choices are correct.
- C. Math.min(z, Math.min(x, y)) ✔ All choices are correct.
- D. Math.min(Math.min(y, z), x) ✔ All choices are correct.

All choices are correct.

All choices are correct.

The correct answers are: Math.min(x, Math.min(y, z)), Math.min(Math.min(x, y), z), Math.min(Math.min(y, z), x), Math.min(z, Math.min(x, y))

Question 38

Correct

Mark 1.00 out of 1.00

Which of the following loops prints "Welcome to Java" 10 times?

A:
```
for (int count = 1; count <= 10; count++) {
  System.out.println("Welcome to Java");
}
```

B:
```
for (int count = 0; count < 10; count++) {
  System.out.println("Welcome to Java");
}
```

C:
```
for (int count = 1; count < 10; count++) {
  System.out.println("Welcome to Java");
}
```

D:
```
for (int count = 0; count <= 10; count++) {
  System.out.println("Welcome to Java");
}
```

Select one:

- A. BD
- B. ABC
- C. AC
- D. BC
- E. AB ✔

In A, the loop displays Welcome to Java 10 times for count from 1 to 10. In B, the loop displays Welcome to Java 10 times for count from 0 to 9. In C, the loop displays Welcome to Java 9 times for count from 1 to 9. In D, the loop displays Welcome to Java 11 times for count from 0 to 10. Therefore, the correct answer is AB.

In A, the loop displays Welcome to Java 10 times for count from 1 to 10. In B, the loop displays Welcome to Java 10 times for count from 0 to 9. In C, the loop displays Welcome to Java 9 times for count from 1 to 9. In D, the loop displays Welcome to Java 11 times for count from 0 to 10. Therefore, the correct answer is AB.

In A, the loop displays Welcome to Java 10 times for count from 1 to 10. In B, the loop displays Welcome to Java 10 times for count from 0 to 9. In C, the loop displays Welcome to Java 9 times for count from 1 to 9. In D, the loop displays Welcome to Java 11 times for count from 0 to 10. Therefore, the correct answer is AB.

The correct answer is: AB

Question 39

Correct

Mark 2.00 out of 2.00

How many times is the println statement executed?

```
for (int i = 0; i < 10; i++)
  for (int j = 0; j < i; j++)
    System.out.println(i * j)
```

Select one:

- A. 45 ✔ When i is 0, the println statement is not executed. When i is 1, the println is executed 1 time. When i is 2, the println is executed 2 times. When i is 9, the println is executed 9 times. So, the total time for the println to be executed is 1 + 2 + ... + 9 = 45.
- B. 20
- C. 100
- D. 10

When i is 0, the println statement is not executed. When i is 1, the println is executed 1 time. When i is 2, the println is executed 2 times. When i is 9, the println is executed 9 times. So, the total time for the println to be executed is 1 + 2 + ... + 9 = 45.

When i is 0, the println statement is not executed. When i is 1, the println is executed 1 time. When i is 2, the println is executed 2 times. When i is 9, the println is executed 9 times. So, the total time for the println to be executed is 1 + 2 + ... + 9 = 45.

The correct answer is: 45

Question 40

Correct

Mark 2.00 out of 2.00

What is the value of balance after the following code is executed?

```
int balance = 10;

while (balance >= 1) {
  if (balance < 9)
    break;
  balance = balance - 9;
}
```

Select one:

- A. -1
- B. 0
- C. 1 ✔ Before the loop, balance is 10. The loop-continuation-condition is true (10 >= 1). In the first iteration, balance is reduced to 1. Since 1 >= 1 is true, the loop body is executed. Since balance < 9 is true, the break statement is executed to exit the loop. So, balance is 1 after the loop is finished. The correct answer for this question is C.
- D. 2

Before the loop, balance is 10. The loop-continuation-condition is true (10 >= 1). In the first iteration, balance is reduced to 1. Since 1 >= 1 is true, the loop body is executed. Since balance < 9 is true, the break statement is executed to exit the loop. So, balance is 1 after the loop is finished. The correct answer for this question is C.

Before the loop, balance is 10. The loop-continuation-condition is true (10 >= 1). In the first iteration, balance is reduced to 1. Since 1 >= 1 is true, the loop body is executed. Since balance < 9 is true, the break statement is executed to exit the loop. So, balance is 1 after the loop is finished. The correct answer for this question is C.

The correct answer is: 1

Question 41

Incorrect

Mark 0.00 out of 2.00

How many times is the println statement executed?

```
for (int i = 0; i < 10; i++)
  for (int j = 0; j < 10; j++)
    System.out.println(i * j);
```

Select one:

- A. 100
- B. 20
- C. 10
- D. 45 ✖

The outer loop is executed 10 times for i from 0 to 9 and the inner loop is executed 10 times from j from 0 to 9. So, the print statement is executed 100 times. The correct answer is A.

The correct answer is: 100

Question 42

Incorrect

Mark 0.00 out of 2.00

What is the value in count after the following loop is executed?

```
int count = 0;
do {
  System.out.println("Welcome to Java");
} while (count++ < 9);
System.out.println(count);
```

Select one:

- A. 8
- B. 9 ✖
- C. 10
- D. 11
- E. 0

The count is initialized to 0 before the loop. The loop is executed for the first time when count is 0. (count++ < 9) increments count by 1 and uses the old count value to check if count < 9. So, the loop is executed 10 times for count from 0 to 9. Note that when count is 9, the loop body is executed and then count++ increments count to 10. The previous count value 9 < 9 is false. So, the loop is finished. Now count becomes 10 after the loop exits. The correct answer is C.

The correct answer is: 10

Question 43

Correct

Mark 1.00 out of 1.00

The _____ method parses a string s to an int value.

Select one:

- A. integer.parseInt(s);
- B. Integer.parseInt(s); ✔ The parseInt method is defined in the Integer class. B is correct.
- C. integer.parseInteger(s);
- D. Integer.parseInteger(s);

The parseInt method is defined in the Integer class. B is correct.

The parseInt method is defined in the Integer class. B is correct.

The correct answer is: Integer.parseInt(s);

Question 44

Correct

Mark 2.00 out of 2.00

The expression "Java " + 1 + 2 + 3 evaluates to _____.

Select one:

- A. Java123
- B. Java6
- C. Java 123 ✔ The + operator is evaluated from left to right. When a string adds with a number, the number is converted into a string. The correct answer is C.
- D. java 123
- E. Illegal expression

The + operator is evaluated from left to right. When a string adds with a number, the number is converted into a string. The correct answer is C.

The + operator is evaluated from left to right. When a string adds with a number, the number is converted into a string. The correct answer is C.

The correct answer is: Java 123

Question 45

Correct

Mark 2.00 out of 2.00

Analyze the following code:

```
int i = 3434; double d = 3434;
System.out.printf("%5.1f %5.1f", i, d);
```

Select one:

- A. The code compiles and runs fine to display 3434.0 3434.0.
- B. The code compiles and runs fine to display 3434 3434.0.
- C. i is an integer, but the format specifier %5.1f specifies a format for double value. The code has an error. ✔ i is an integer, but the format specifier %5.1f specifies a format for double value. Type does not match. So, the correct answer is C.

i is an integer, but the format specifier %5.1f specifies a format for double value. Type does not match. So, the correct answer is C.

i is an integer, but the format specifier %5.1f specifies a format for double value. Type does not match. So, the correct answer is C.

The correct answer is: i is an integer, but the format specifier %5.1f specifies a format for double value. The code has an error.

Question 46

Correct

Mark 1.00 out of 1.00

What is the number of iterations in the following loop?

```
for (int i = 1; i <= n; i++) {
  // iteration
}
```

Select one:

- A. 2*n
- B. n ✔ The loop is executed n times for i from 1 to n. So, the correct answer is B.
- C. n - 1
- D. n + 1

The loop is executed n times for i from 1 to n. So, the correct answer is B.

The loop is executed n times for i from 1 to n. So, the correct answer is B.

The correct answer is: n

Question 47

Incorrect

Mark 0.00 out of 2.00

"AbA".compareToIgnoreCase("abC") returns _____.

Select one:

- A. 1 ✖
- B. 2
- C. -1
- D. -2
- E. 0

Ignoring case, you compare aba with abc. The first two characters in the two strings are the same. The different between the last two characters is -2. The correct answer is D.

The correct answer is: -2

Question 48

Correct

Mark 2.00 out of 2.00

Math.cos(Math.PI) returns _____.

Select one:

- A. 1.0
- B. 0.0
- C. 0.5
- D. -1.0 ✔ Note that Math.PI is 180 degrees.

Note that Math.PI is 180 degrees.

Note that Math.PI is 180 degrees.

The correct answer is: -1.0

Question 49

Partially correct

Mark 1.00 out of 2.00

Which of the following is not a correct method in the Character class? Please select all that apply.

Select one or more:

- A. isLetterOrDigit(char) ✖
- B. toLowerCase(char) ✖
- C. isDigit()
- D. isLetter(char)
- E. toUpperCase() ✔ isDigit() should be isDigit(char) and toUpperCase() should be toUpperCase(char)

isDigit() should be isDigit(char) and toUpperCase() should be toUpperCase(char)

The correct answers are: isDigit(), toUpperCase()

Question 50

Correct

Mark 2.00 out of 2.00

Which of the following is an invalid specifier for the printf statement?

Select one:

- A. %4c
- B. %10b
- C. %6d
- D. %8.2d ✔ All correct except D. D is wrong because the specifier d is for decimal integer.
- E. %10.2e

All correct except D. D is wrong because the specifier d is for decimal integer.

All correct except D. D is wrong because the specifier d is for decimal integer.

The correct answer is: %8.2d

Question 51

Correct

Mark 2.00 out of 2.00

Analyze the following code. Please select all that apply.

```
int count = 0;
while (count < 100) {
 // Point A
 System.out.println("Welcome to Java!");
 count++;
 // Point B
}

// Point C
```

Select one or more:

- A. count < 100 is always true at Point A ✔ The count is initialized to 0 before the loop. The loop is executed 100 times for count from 0 to 99. Inside the loop body, at Point A, count < 100 is true. After executing count++, count < 100 in Point B may become false. After the loop is exited, count < 100 is false at Point C. So, the correct answer is AE.
- B. count < 100 is always true at Point B
- C. count < 100 is always false at Point B
- D. count < 100 is always true at Point C
- E. count < 100 is always false at Point C ✔ The count is initialized to 0 before the loop. The loop is executed 100 times for count from 0 to 99. Inside the loop body, at Point A, count < 100 is true. After executing count++, count < 100 in Point B may become false. After the loop is exited, count < 100 is false at Point C. So, the correct answer is AE.

The count is initialized to 0 before the loop. The loop is executed 100 times for count from 0 to 99. Inside the loop body, at Point A, count < 100 is true. After executing count++, count < 100 in Point B may become false. After the loop is exited, count < 100 is false at Point C. So, the correct answer is AE.

The count is initialized to 0 before the loop. The loop is executed 100 times for count from 0 to 99. Inside the loop body, at Point A, count < 100 is true. After executing count++, count < 100 in Point B may become false. After the loop is exited, count < 100 is false at Point C. So, the correct answer is AE.

The correct answers are: count < 100 is always true at Point A, count < 100 is always false at Point C

Question 52

Correct

Mark 2.00 out of 2.00

Analyze the following code:

```
import java.util.Scanner;

public class Test {
  public static void main(String[] args) {
    int sum = 0;
    for (int i = 0; i < 100000; i++) {
      Scanner input = new Scanner(System.in);
      sum += input.nextInt();
    }
  }
}
```

Select one:

- A. The program does not compile because the Scanner input = new Scanner(System.in); statement is inside the loop.
- B. The program compiles, but does not run because the Scanner input = new Scanner(System.in); statement is inside the loop.
- C. The program compiles and runs, but it is not efficient and unnecessary to execute the Scanner input = new Scanner(System.in); statement inside the loop. You should move the statement before the loop. ✔ To receive input from the keyboard, you need to create an input object from the Scanner class. You should create this object only once in the program. Placing the statement Scanner input = new Scanner(System.in) in the loop causes it to be created multiple times, which is a bad practice and could lead to potential errors. So, the correct answer is C.
- D. The program compiles, but does not run because there is not prompting message for entering the input.

To receive input from the keyboard, you need to create an input object from the Scanner class. You should create this object only once in the program. Placing the statement Scanner input = new Scanner(System.in) in the loop causes it to be created multiple times, which is a bad practice and could lead to potential errors. So, the correct answer is C.

To receive input from the keyboard, you need to create an input object from the Scanner class. You should create this object only once in the program. Placing the statement Scanner input = new Scanner(System.in) in the loop causes it to be created multiple times, which is a bad practice and could lead to potential errors. So, the correct answer is C.

The correct answer is: The program compiles and runs, but it is not efficient and unnecessary to execute the Scanner input = new Scanner(System.in); statement inside the loop. You should move the statement before the loop.

Question 53

Correct

Mark 2.00 out of 2.00

Which of the following is correct to obtain the square root of 5? Please select all that apply.

Select one or more:

- A. Math.sqrt(5) ✔ Math.sqrt(5) is the same as Math.pow(5, 0.5), resulting the square root of 5.
- B. Math.pow(5, 0.5) ✔ Math.sqrt(5) is the same as Math.pow(5, 0.5), resulting the square root of 5.
- C. Math.sqrt(25)
- D. Math.pow(5, 2)

Math.sqrt(5) is the same as Math.pow(5, 0.5), resulting the square root of 5.

Math.sqrt(5) is the same as Math.pow(5, 0.5), resulting the square root of 5.

The correct answers are: Math.sqrt(5), Math.pow(5, 0.5)

Question 54

Correct

Mark 2.00 out of 2.00

What will be displayed when the following code is executed?

```
int number = 6;
while (number > 0) {
  number -= 3;
  System.out.print(number + " ");
}
```

Select one:

- A. 6 3 0
- B. 6 3
- C. 3 0 ✔ number is 6 before the loop. In the first iteration, number is reduced to 3. In the second iteration, number is reduced to 0. The loop is now finished. The loop body is executed 2 times for number 6 and 3. Since number is reduced by 3 before the print statement. 3 and 0 and displayed. So, the correct answer is C.
- D. 3 0 -3
- E. 0 -3

number is 6 before the loop. In the first iteration, number is reduced to 3. In the second iteration, number is reduced to 0. The loop is now finished. The loop body is executed 2 times for number 6 and 3. Since number is reduced by 3 before the print statement. 3 and 0 and displayed. So, the correct answer is C.

number is 6 before the loop. In the first iteration, number is reduced to 3. In the second iteration, number is reduced to 0. The loop is now finished. The loop body is executed 2 times for number 6 and 3. Since number is reduced by 3 before the print statement. 3 and 0 and displayed. So, the correct answer is C.

The correct answer is: 3 0

Question 55

Correct

Mark 2.00 out of 2.00

Is the following loop correct?

```
for ( ; ; );
```

Select one:

- A. Yes ✔ Yes. This is equivalent to for (; true;).
- B. No

Yes. This is equivalent to for (; true;).

Yes. This is equivalent to for (; true;).

The correct answer is: Yes

Question 56

Correct

Mark 1.00 out of 1.00

The statement System.out.printf("%5d", 123456) outputs _____.

Select one:

- A. 12345
- B. 23456
- C. 123456 ✔ %5d specifies an integer with width 5. The width is automatically expanded if the number is larger than the specified width. So, the correct answer is C.
- D. 12345.6

%5d specifies an integer with width 5. The width is automatically expanded if the number is larger than the specified width. So, the correct answer is C.

%5d specifies an integer with width 5. The width is automatically expanded if the number is larger than the specified width. So, the correct answer is C.

The correct answer is: 123456

Question 57

Partially correct

Mark 0.50 out of 1.00

Suppose s1 and s2 are two strings. Which of the following statements or expressions is incorrect? Please select all that apply.

Select one or more:

- A. String s3 = s1 - s2; ✔ A is wrong because the - operator cannot be used for strings. B is wrong because the compareTo method returns an int, not a boolean. C is wrong because the [] cannot be used for accessing string elements. D is wrong because of index out of bounds.
- B. boolean b = s1.compareTo(s2); ✔ A is wrong because the - operator cannot be used for strings. B is wrong because the compareTo method returns an int, not a boolean. C is wrong because the [] cannot be used for accessing string elements. D is wrong because of index out of bounds.
- C. char c = s1[0];
- D. char c = s1.charAt(s1.length());

A is wrong because the - operator cannot be used for strings. B is wrong because the compareTo method returns an int, not a boolean. C is wrong because the [] cannot be used for accessing string elements. D is wrong because of index out of bounds.

The correct answers are: String s3 = s1 - s2;, boolean b = s1.compareTo(s2);, char c = s1[0];, char c = s1.charAt(s1.length());